

DFS.c - Code::Blocks 25.03

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<global>main(): int

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Start here x BFS.c x DFS.c x

```
1  #include <stdio.h>
2  #define MAX 100
3
4  int visited[MAX];
5  void DFS(int adj[MAX][MAX], int n, int v) {
6      visited[v] = 1;
7
8      for (int i = 0; i < n; i++) {
9          if (adj[v][i] == 1 && !visited[i]) {
10             DFS(adj, n, i);
11         }
12     }
13 }
14 int main() {
15     int n, adj[MAX][MAX];
16
17     printf("Enter number of vertices: ");
18     scanf("%d", &n);
19
20     printf("Enter adjacency matrix:\n");
21     for (int i = 0; i < n; i++) {
22         for (int j = 0; j < n; j++) {
23             scanf("%d", &adj[i][j]);
24         }
25     }
26     for (int i = 0; i < n; i++) {
27         visited[i] = 0;
28     }
29     DFS(adj, n, 0);
30
31     int connected = 1;
32     for (int i = 0; i < n; i++) {
33         if (!visited[i]) {
34             connected = 0;
35             break;
36         }
37     }
38     if (connected)
39         printf("The graph is CONNECTED.\n");
40     else
41         printf("The graph is NOT CONNECTED.\n");
42     return 0;
43 }
44 }
```

C:\Users\BMSCECSE-L4\Desktop\1BM24C5173\DFS.cC/C++Windows (CR+LF)WINDOWS-1252Line 41, Col 49, Pos 906InsertRead/Write default

22°C Sunny

Search

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```
Enter number of vertices: 6
Enter adjacency matrix:
0 1 1 0 0 0
1 0 0 0 0 1
1 0 0 1 0 1
0 0 1 0 1 1
0 0 0 1 0 1
0 1 1 1 1 0
The graph is CONNECTED.

Process returned 0 (0x0)   execution time : 62.111 s
Press any key to continue.
```

```
Enter number of vertices: 5
Enter adjacency matrix:
0 1 0 0 0
1 0 0 0 0
0 0 0 1 1
0 0 1 0 0
0 0 1 0 0
The graph is NOT CONNECTED.

Process returned 0 (0x0)   execution time : 48.432 s
Press any key to continue.
|
```